

PRODUCT DESIGN CASE STUDY

Personalized Learning Recommendation System

Designed for Learners, Instructors, and Platform Admins

+25% engagement · +30% completion · +20% satisfaction

Prepared for: Airtribe — Tech 101 | Core Project Assignment

Three Personas, One Recommendation Engine

Every design decision in this deck is evaluated against these three user types

Learners

Discover relevant courses matched to interests, skill level, and career goals

Instructors / Content Creators

Understand demand, optimize course metadata, and reach the right learners

Platform Admins

Monitor system health, configure business rules, and ensure compliance

Learners

“As a learner, I want to [need] so that I can [benefit].”

1 See course recommendations based on my career goals, so I can find relevant learning paths faster

2 Set my skill level for each topic, so recommended courses match my current ability

3 See why a course was recommended to me, so I can trust the suggestion

4 Save courses to a wishlist, so I can revisit them later

5 Get notified when a new course matching my interests is released, so I stay engaged

Instructors & Platform Admins

INSTRUCTORS / CONTENT CREATORS

See which topics learners search for but can't find courses on, so I can create content that fills the gap

See how my course performs in the recommendation engine, so I can optimize its metadata

Tag my course with accurate skill-level and prerequisite metadata, so it reaches the right learners

View engagement analytics on recommended vs. searched enrollments, so I understand what drives sign-ups

PLATFORM ADMINS

Monitor the overall health of the recommendation engine, so I can catch performance or bias issues early

Configure business rules (e.g., promote new courses), so I can balance personalization with strategic goals

Audit recommendation logs for compliance, so we meet data protection regulations

A/B test different recommendation algorithms, so I can measure impact on completion rates

10 Core Features

User Type	Feature	Description	Priority
Learner	Personalized Course Feed	Home feed ranked by relevance to interests, skill, and goals	High
Learner	Skill Assessment Quiz	Short onboarding quiz calibrating skill level per topic	High
Learner	“Why This Course?” Explainability	Shows the reason behind each recommendation	Medium
Learner	Next-Course Suggestion	Recommends the next course upon completion	High
Instructor	Content Gap Insights	Surfaces searched-but-unfulfilled topics	Medium
Instructor	Course Performance Analytics	Impressions, CTR, and enrollment source per course	High
Instructor	Metadata Tagging Tool	Structured tagging for skill level, prerequisites, outcomes	High
Admin	Recommendation Engine Health Dashboard	Monitors latency, coverage, diversity, and drift	High
Admin	Business Rules Configuration	Set boosts/demotions to balance personalization with strategy	Medium
Admin	A/B Testing Framework	Run experiments across ranking algorithms	Medium

System Qualities

1–2 requirements per category

Performance

- Recommendations load within 300ms (p95) on the home feed
- Re-ranking completes within 1s of a new completion event

Scalability

- Supports 10M+ learners and 100K+ courses without added latency
- Scales horizontally to absorb 5x traffic during enrollment campaigns

Availability

- 99.9% uptime for the recommendation service
- Graceful fallback to popularity-based ranking if personalization is down

Compliance

- Complies with GDPR for behavioral data used in personalization
- Explicit consent management for data used in recommendations

Security & Data Retention

- Behavioral data encrypted at rest and in transit
- Raw interaction logs retained 18 months max, then anonymized

Monitoring & Logging

- Track impressions, CTR, and conversion per model version
- Alert on model drift or sudden engagement drops

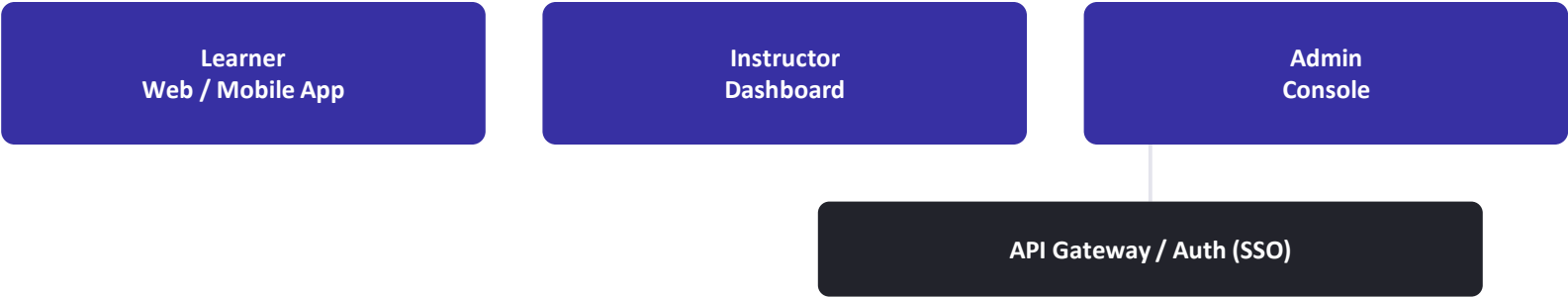
Extensibility

- Architecture supports new signals (e.g., career-goal embeddings) without a full rebuild
- API-first design so new surfaces (mobile, LMS) can query recommendations

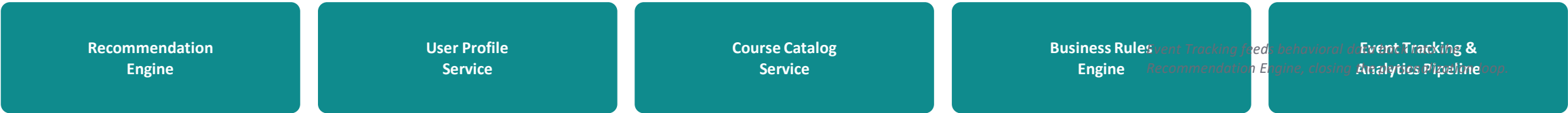
Block Diagram

User interface, backend system, and platform integration

USER INTERFACE



BACKEND SYSTEM



PLATFORM INTEGRATION



Personalization vs. Privacy

The major trade-off in building this system

PERSONALIZATION

More behavioral tracking — clicks, time-on-page, search queries — improves recommendation accuracy and relevance.

vs.

PRIVACY

Deeper tracking increases privacy risk, data-retention burden, and regulatory exposure (GDPR, data protection).

HOW WE'D PRIORITIZE

As an education platform, trust and compliance carry a higher bar than most consumer apps. We'd favor privacy by default: rely primarily on explicit, low-risk signals (stated goals, skill assessment, completions) rather than deep implicit tracking.

Product decision: Adopt a consent-tiered personalization model. A baseline experience for all users runs on explicit, low-risk signals. Users can opt into an “enhanced personalization” tier that uses richer behavioral tracking, with clear communication of the value exchange — preserving both trust and recommendation quality.

Key Assumptions & What's Next

ASSUMPTIONS

- Users already have profiles with basic demographic and career-goal info at onboarding
- Course catalog metadata (skill level, prerequisites, tags) is reasonably well-maintained by instructors
- Enough historical interaction data exists, or can be bootstrapped with content-based filtering
- Learners are willing to complete a short skill assessment during onboarding
- Instructors are incentivized (via performance dashboards) to keep metadata accurate

FUTURE PLANNING

- Incorporate career-path/job-outcome data to tie recommendations to real career transitions
- Add peer-cohort recommendations (“learners like you also took...”) once data density supports it
- Expand explainability with interactive controls (e.g., adjust for skill vs. speed vs. depth)
- Explore on-device or federated personalization to reduce privacy exposure as usage scales
- Extend the engine to instructors — suggest content gaps based on aggregated learner demand

Summary

14

user stories across 3 personas

10

core features prioritized

7

non-functional requirement categories covered

Strategic thread

- The system is designed around three personas whose needs reinforce each other — better instructor metadata improves learner recommendations, which improves admin-visible engagement metrics.
- Privacy and personalization are treated as a deliberate, consent-based trade-off rather than an afterthought — appropriate for an education platform's trust bar.
- The architecture is built to extend — new signals, new personas, and new consumption surfaces can be added without a full rebuild.